

PE-EC702C — Neural Network & Fuzzy Logic Control

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Module 4 Exam Notes: Fuzzy Sets and Membership Functions

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1. Classical (Crisp) Sets vs. Fuzzy Sets

Classical set theory is based on binary logic (true/false, black/white), whereas fuzzy set theory allows for gradual transitions and degrees of membership.

Fuzzy Set

A **Fuzzy Set** A in a universe of discourse U is defined by a membership function $\mu_A(x)$ that maps each element $x \in U$ to a continuous value in the interval $[0, 1]$.

$$A = \{(x, \mu_A(x)) \mid x \in U\} \quad (1)$$

1.1 Standard Fuzzy Operations

Let A and B be two fuzzy sets with membership functions $\mu_A(x)$ and $\mu_B(x)$ respectively:

- **Union (OR / Standard Max):**

$$\mu_{A \cup B}(x) = \max(\mu_A(x), \mu_B(x)) = \mu_A(x) \vee \mu_B(x) \quad (2)$$

- **Intersection (AND / Standard Min):**

$$\mu_{A \cap B}(x) = \min(\mu_A(x), \mu_B(x)) = \mu_A(x) \wedge \mu_B(x) \quad (3)$$

- **Complement (NOT):**

$$\mu_{\bar{A}}(x) = 1 - \mu_A(x) \quad (4)$$

- **Algebraic Product (Dot):** $\mu_{A \cdot B}(x) = \mu_A(x) \cdot \mu_B(x)$

- **Algebraic Sum (Probabilistic sum):** $\mu_{A \hat{+} B}(x) = \mu_A(x) + \mu_B(x) - \mu_A(x)\mu_B(x)$

2. Fuzzy Relations and Composition

Fuzzy relations map elements from universe X to universe Y with a membership value in $[0, 1]$.

Fuzzy Compositions

Let R be a fuzzy relation from X to Y , and S be a fuzzy relation from Y to Z .

1. **Max-Min Composition** ($T = R \circ S$):

$$\mu_T(x, z) = \max_{y \in Y} \{\min(\mu_R(x, y), \mu_S(y, z))\} \quad (5)$$

2. **Max-Product Composition** ($T = R \bullet S$):

$$\mu_T(x, z) = \max_{y \in Y} \{\mu_R(x, y) \cdot \mu_S(y, z)\} \quad (6)$$

2.1 Tolerance and Equivalence Relations

Fuzzy relations can possess algebraic properties similar to classical relations:

- **Reflexivity:** A relation R on set X is reflexive if the membership of any element paired with itself is 1:

$$\mu_R(x, x) = 1 \quad \forall x \in X \quad (7)$$

- **Symmetry:** A relation R is symmetric if order of elements does not affect membership:

$$\mu_R(x, y) = \mu_R(y, x) \quad \forall x, y \in X \quad (8)$$

- **Transitivity (Max-Min Transitivity):** A relation R is Max-Min transitive if:

$$R \circ R \subseteq R \quad \implies \quad \mu_R(x, z) \geq \max_{y \in X} \{\min(\mu_R(x, y), \mu_R(y, z))\} \quad \forall x, z \in X \quad (9)$$

Based on these properties, we define two special fuzzy relations:

1. **Tolerance Relation (Compatibility Relation):** A fuzzy relation that is **reflexive** and **symmetric**, but not necessarily transitive. It measures the degree of similarity or compatibility between elements.
2. **Equivalence Relation (Similarity Relation):** A fuzzy relation that is **reflexive**, **symmetric**, and **Max-Min transitive**. It generalizes the classical equivalence relation to group elements into fuzzy equivalence classes.

3. Membership Functions and Standard Forms

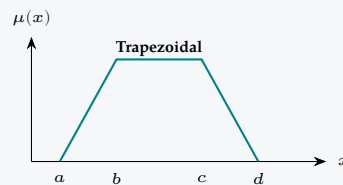
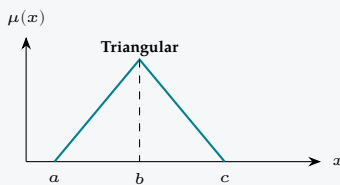
3.1 Key Geometric Features

- **Support:** The crisp set containing all elements with non-zero membership: $\{x \in U \mid \mu_A(x) > 0\}$.
- **Core:** The crisp set containing all elements with full membership: $\{x \in U \mid \mu_A(x) = 1\}$.
- **Boundary:** The region with $0 < \mu_A(x) < 1$.
- **Crossover Points:** The elements where membership is exactly 0.5: $\{x \in U \mid \mu_A(x) = 0.5\}$.

3.2 Standard Shapes

Standard shapes are used to represent categories like "Warm Temperature" or "High Voltage":

Standard Membership Function Shapes



4. Fuzzification and Defuzzification

- **Fuzzification:** The process of converting a crisp physical input value into a fuzzy membership value using predefined membership functions.
- **Defuzzification:** The process of converting a combined fuzzy output shape into a single representative crisp output value.

4.1 Standard Defuzzification Methods

Let C be the output fuzzy set defined on the continuous output space z .

Mathematical Formulations of Defuzzification

1. Centroid (Center of Gravity - COG):

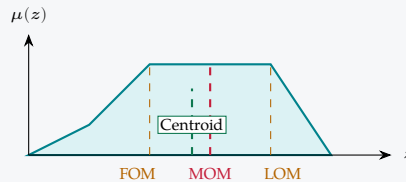
$$z^* = \frac{\int z \mu_C(z) dz}{\int \mu_C(z) dz} \rightarrow \text{(Discrete: } z^* = \frac{\sum z_i \mu_C(z_i)}{\sum \mu_C(z_i)}) \quad (10)$$

2. Mean of Maxima (MOM):

$$z^* = \frac{\sum_{z_i \in M} z_i}{|M|} \quad \text{where } M = \{z \mid \mu_C(z) \text{ is maximum}\} \quad (11)$$

3. First of Maxima (FOM) / Last of Maxima (LOM): $z^* = \min(M)$ (First) or $z^* = \max(M)$ (Last).

Defuzzification Geometric Points



5. Lambda Cuts (λ -cuts)

A λ -cut (or α -cut) operation converts a fuzzy set or fuzzy relation into a classical (crisp) set by thresholding.

Lambda Cut (λ -cut)

The λ -cut of a fuzzy set A , denoted by A_λ , is a crisp set containing all elements whose membership values are greater than or equal to λ :

$$A_\lambda = \{x \in U \mid \mu_A(x) \geq \lambda\} \quad \text{where } \lambda \in [0, 1] \quad (12)$$

6. Worked Examples

Fuzzy Set Operations

Problem: Given two discrete fuzzy sets defined on $U = \{1, 2, 3, 4\}$:

$$A = \left\{ \frac{0.2}{1}, \frac{0.7}{2}, \frac{1.0}{3}, \frac{0.4}{4} \right\}, \quad B = \left\{ \frac{0.5}{1}, \frac{0.6}{2}, \frac{0.3}{3}, \frac{0.9}{4} \right\}$$

Calculate: (a) $A \cup B$, (b) $A \cap B$, and (c) the algebraic product $A \cdot B$.

Solution: Recall: $\mu_{A \cup B} = \max(\mu_A, \mu_B)$, $\mu_{A \cap B} = \min(\mu_A, \mu_B)$, and $\mu_{A \cdot B} = \mu_A \cdot \mu_B$.

• (a) **Union ($A \cup B$):**

$$\mu_{A \cup B}(1) = \max(0.2, 0.5) = 0.5$$

$$\mu_{A \cup B}(2) = \max(0.7, 0.6) = 0.7$$

$$\mu_{A \cup B}(3) = \max(1.0, 0.3) = 1.0$$

$$\mu_{A \cup B}(4) = \max(0.4, 0.9) = 0.9$$

$$A \cup B = \left\{ \frac{0.5}{1}, \frac{0.7}{2}, \frac{1.0}{3}, \frac{0.9}{4} \right\}$$

• (b) **Intersection ($A \cap B$):**

$$\begin{aligned} A \cap B &= \left\{ \frac{\min(0.2, 0.5)}{1}, \frac{\min(0.7, 0.6)}{2}, \frac{\min(1.0, 0.3)}{3}, \frac{\min(0.4, 0.9)}{4} \right\} \\ &= \left\{ \frac{0.2}{1}, \frac{0.6}{2}, \frac{0.3}{3}, \frac{0.4}{4} \right\} \end{aligned}$$

• (c) **Algebraic Product ($A \cdot B$):**

$$\begin{aligned} A \cdot B &= \left\{ \frac{0.2 \times 0.5}{1}, \frac{0.7 \times 0.6}{2}, \frac{1.0 \times 0.3}{3}, \frac{0.4 \times 0.9}{4} \right\} \\ &= \left\{ \frac{0.10}{1}, \frac{0.42}{2}, \frac{0.30}{3}, \frac{0.36}{4} \right\} \end{aligned}$$

Max-Min Composition of Fuzzy Relations

Problem: Given fuzzy relations R (size 2×2) and S (size 2×2):

$$R = \begin{bmatrix} 0.6 & 0.3 \\ 0.2 & 0.9 \end{bmatrix}, \quad S = \begin{bmatrix} 0.5 & 0.8 \\ 0.7 & 0.1 \end{bmatrix}$$

Compute the Max-Min composition $T = R \circ S$.

Solution: The formulation is $t_{ik} = \max_j \{ \min(r_{ij}, s_{jk}) \}$.

• **Element t_{11} :**

$$t_{11} = \max(\min(0.6, 0.5), \min(0.3, 0.7)) = \max(0.5, 0.3) = 0.5$$

- **Element t_{12} :**

$$t_{12} = \max(\min(0.6, 0.8), \min(0.3, 0.1)) = \max(0.6, 0.1) = 0.6$$

- **Element t_{21} :**

$$t_{21} = \max(\min(0.2, 0.5), \min(0.9, 0.7)) = \max(0.2, 0.7) = 0.7$$

- **Element t_{22} :**

$$t_{22} = \max(\min(0.2, 0.8), \min(0.9, 0.1)) = \max(0.2, 0.1) = 0.2$$

Thus, the composition is:

$$T = R \circ S = \begin{bmatrix} 0.5 & 0.6 \\ 0.7 & 0.2 \end{bmatrix}$$

Discrete Centroid Defuzzification

Problem: Calculate the crisp value z^* of the fuzzy set C using Centroid defuzzification:

$$C = \left\{ \frac{0.1}{10}, \frac{0.3}{20}, \frac{0.8}{30}, \frac{1.0}{40}, \frac{0.6}{50} \right\}$$

Solution: We use the discrete Centroid formula: $z^* = \frac{\sum z_i \mu_C(z_i)}{\sum \mu_C(z_i)}$.

$$\begin{aligned} \sum z_i \mu_C(z_i) &= 10(0.1) + 20(0.3) + 30(0.8) + 40(1.0) + 50(0.6) \\ &= 1 + 6 + 24 + 40 + 30 = 101 \end{aligned}$$

$$\begin{aligned} \sum \mu_C(z_i) &= 0.1 + 0.3 + 0.8 + 1.0 + 0.6 = 2.8 \\ z^* &= \frac{101}{2.8} \approx 36.07 \end{aligned}$$

The defuzzified crisp output is $z^* \approx 36.07$.

Quick Revision — Module IV

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Key Formulas at a Glance

Fuzzy Union (OR)	$\mu_{A \cup B}(x) = \max(\mu_A(x), \mu_B(x))$
Fuzzy Intersection (AND)	$\mu_{A \cap B}(x) = \min(\mu_A(x), \mu_B(x))$
Max-Min Composition	$\mu_{R \circ S}(x, z) = \max_y \{ \min(\mu_R(x, y), \mu_S(y, z)) \}$
Max-Product Composition	$\mu_{R \bullet S}(x, z) = \max_y \{ \mu_R(x, y) \cdot \mu_S(y, z) \}$
Centroid Defuzzification	$z^* = \frac{\sum z_i \mu_C(z_i)}{\sum \mu_C(z_i)} \quad \text{or} \quad \frac{\int z \mu_C(z) dz}{\int \mu_C(z) dz}$
Lambda-Cut (λ-cut)	$A_\lambda = \{x \in U \mid \mu_A(x) \geq \lambda\}$

One-Line Definitions

- **Fuzzy Set:** A set containing elements with varying degrees of membership represented by values in $[0, 1]$.
- **Membership Function:** The graphical curve or mathematical function that maps elements to their membership degrees.
- **Centroid Defuzzification:** A defuzzification method that calculates the center of gravity of the output fuzzy region.
- **Lambda Cut:** An operation that converts a fuzzy set/relation into a classical set using a threshold value λ .
- **Crossover Point:** Any element in the universe of discourse whose membership degree in a fuzzy set is exactly 0.5.

Frequently Asked / Viva Questions

1. What is the difference between probability and fuzziness?

Probability deals with the likelihood/chance of an event occurring (undecided outcome). Fuzziness deals with the vagueness of an event that has already occurred (deterministic but vague boundaries, e.g. "He is tall").

2. State the properties of classical sets that do NOT hold in fuzzy sets.

The Law of Contradiction ($A \cap \bar{A} = \emptyset$) and the Law of Excluded Middle ($A \cup \bar{A} = U$) do not hold in fuzzy sets (e.g. $A \cap \bar{A} \neq \emptyset$).

3. What is a normal fuzzy set?

A fuzzy set is normal if its height is exactly 1.0 (i.e. at least one element has a membership value of 1).

4. Define a subnormal fuzzy set.

A fuzzy set whose maximum membership value (height) is strictly less than 1.0.

Frequently Asked / Viva Questions (contd.)**5. Why is Centroid defuzzification the most commonly used method?**

Because it integrates the entire area of the output fuzzy shape, yielding smooth, continuous output transitions relative to changes in inputs.

6. Explain the Max-Min composition rule.

It is a method of relating elements of set X to set Z through intermediate set Y . For each (x, z) pair, it finds the strength of paths through all $y \in Y$ (using Min) and selects the strongest path (using Max).

7. What is the difference between a tolerance relation and an equivalence relation?

An equivalence relation must be reflexive, symmetric, and transitive. A tolerance relation is only reflexive and symmetric (not transitive).

8. Define the support of a fuzzy set.

The crisp set of all elements in the universe of discourse whose membership values are strictly greater than 0 ($\mu_A(x) > 0$).

9. What are the standard forms of membership functions?

Triangular, Trapezoidal, Gaussian, Sigmoidal, L-shaped, and S-shaped curves.

10. What is the purpose of defuzzification in fuzzy controllers?

Physical actuators and microcontrollers require crisp, exact values (like voltage or valve position) to operate, which requires defuzzification of the fuzzy control decision.

Crisp Sets vs. Fuzzy Sets

Attribute	Classical (Crisp) Sets	Fuzzy Sets
Membership Values	Binary: $\{0, 1\}$	Continuous: $[0, 1]$
Boundaries	Sharp, distinct boundaries	Gradual, vague boundaries
Operations	Boolean logic (AND, OR, NOT)	Fuzzy logic (Min, Max, $1 - \mu$)
Contradiction Law	Holds: $A \cap \bar{A} = \emptyset$	Does not hold: $A \cap \bar{A} \neq \emptyset$